Dynamic Panel Models

BUSS975: Causal Inference in Financial Research

Professor Ji-Woong Chung

Korea University

Where this lecture sits

- Fixed effects removes everything permanent about a firm. The price: *strict exogeneity*, no regressor may respond to past shocks.
- The regressor that responds to past shocks by construction: the outcome's own lag. Leverage, payout, cash, investment and bank capital all depend on their own past.

1. Why the Lagged Outcome Is Endogenous

The model

$$Y_{it} \;=\; \alpha\, Y_{i,t-1} + X_{it}'\beta + v_{it}, \qquad v_{it} = \mu_i + \varepsilon_{it}$$

- α**: state dependence.** In partial-adjustment models (leverage targets, dividends, cash, bank capital), $1-\alpha$ is the speed of adjustment.
- μ_i**: the firm being itself.** Permanent; in leverage it explains most of the variation (Lemmon, Roberts and Zender 2008).
- ε_{it}**: news.** Mean zero, serially uncorrelated, unpredictable from anything known the year before.

- $Y_{i,t-1}$ is built from μ_i at every past date, so pooled OLS, which leaves μ_i in the error, is biased **up**. Last week's remedy: transform μ_i away, by demeaning or by differencing. Does that settle it?

Within estimator: demeaning borrows from the future

Observations $t = 1, \ldots, T$, so Y_{i0} is the first lag. Subtract each firm's time average from every variable:

$$(Y_{it} - \bar{Y}_i) \;=\; \alpha\,(Y_{i,t-1} - \bar{Y}_{i,-1}) + (\varepsilon_{it} - \bar{\varepsilon}_i),$$

$$\bar{Y}_i = \tfrac{1}{T}\textstyle\sum_{s=1}^{T} Y_{is}, \qquad \bar{Y}_{i,-1} = \tfrac{1}{T}\textstyle\sum_{s=1}^{T} Y_{i,s-1}, \qquad \bar{\varepsilon}_i = \tfrac{1}{T}\textstyle\sum_{s=1}^{T} \varepsilon_{is}.$$

- μ_i is gone. But for $t \geq 2$, $\bar{\varepsilon}_i$ contains $\frac{1}{T}\varepsilon_{i,t-1}$, so the demeaned error contains $-\frac{1}{T}\varepsilon_{i,t-1}$, while the regressor $Y_{i,t-1}$ contains $+\varepsilon_{i,t-1}$:

$$\mathrm{Cov}(Y_{i,t-1},\; \varepsilon_{it} - \bar{\varepsilon}_i) \;=\; -\frac{\sigma_\varepsilon^2}{T} \;<\; 0 \;\Longrightarrow\; \hat{\alpha}_{\mathrm{FE}} \text{ biased } \textbf{down}.$$

- A good shock last year raised $Y_{i,t-1}$ *and* raised the mean subtracted from every error. The regressor is high exactly when the transformed error has been pushed down. Demeaning uses the firm's whole history, future years included.

First differencing: the overlap shrinks to one date

$$\Delta Y_{it} \;=\; \alpha\, \Delta Y_{i,t-1} + \Delta \varepsilon_{it}$$

μ_i is gone again. Every remaining object is a bundle of dated shocks; write down which dates are inside each:

Object	shocks dated	shares a date with $\Delta\varepsilon_{it}$?
error $\Delta\varepsilon_{it} = \varepsilon_{it} - \varepsilon_{i,t-1}$	$t,\ t-1$	
regressor $\Delta Y_{i,t-1} = Y_{i,t-1} - Y_{i,t-2}$	$t-1,\ t-2,\ \ldots$	**yes**: $t-1$
$Y_{i,t-2}$	$t-2,\ t-3,\ \ldots$	no
$\Delta Y_{i,t-2}$	$t-2,\ t-3,\ \ldots$	no

- $\mathrm{Cov}(\Delta Y_{i,t-1}, \Delta\varepsilon_{it}) = -\mathrm{Var}(\varepsilon_{i,t-1}) = -\sigma_\varepsilon^2$: a shock at $t-1$ raises the regressor and lowers the error. Biased **down**; more years do not help.
- Not the fix, but **where the fix starts**: μ_i is gone and only a *one-date* overlap remains. The last two rows share no date with the error.

2. Internal Instruments

The assumption we can afford: sequential exogeneity

- **Strict exogeneity**: shocks unrelated to every regressor at *all* dates, past and future,

$$\mathbb{E}\left[\varepsilon_{it} \,\middle|\, X_{i1}, \ldots, X_{iT}, \mu_i\right] = 0 \quad \text{(and likewise for } Y_{i,t-1}, \ldots, Y_{i,T-1}\text{)}.$$

  Fails by construction: ε_{it} moves Y_{it}, tomorrow's regressor.

- **Sequential exogeneity**: shocks unrelated to the *past* of Y and the *past and present* of X,

$$\mathbb{E}\left[\varepsilon_{it} \,\middle|\, Y_{i,t-1}, \ldots, Y_{i1},\ X_{it}, \ldots, X_{i1},\ \mu_i\right] = 0.$$

  ε_{it} **is news**: it may change everything *after* it (feedback allowed) but cannot be forecast from what was known *before* it, that is, ε_{it} is serially uncorrelated.

- Everything dated $t-2$ or earlier was known before the news in $\Delta\varepsilon_{it}$ arrived:

$$\mathbb{E}\left[\, Y_{i,t-s}\, \Delta\varepsilon_{it} \,\right] = 0, \qquad s \geq 2.$$

  These **moment conditions** are the basis of what follows.

Anderson and Hsiao (1981): the first fix

Difference away μ_i, then instrument the one bad regressor, $\Delta Y_{i,t-1}$, by **2SLS**:

$$\Delta Y_{it} = \alpha \, \Delta Y_{i,t-1} + \Delta X_{it}'\beta + \Delta \varepsilon_{it}.$$

instrument z_{it}	relevant because	valid because	equations
level $Y_{i,t-2}$	it is part of $\Delta Y_{i,t-1} = Y_{i,t-1} - Y_{i,t-2}$	dated $t-2$: $\mathbb{E}[Y_{i,t-2}\Delta\varepsilon_{it}] = 0$	$t \geq 2$
difference $\Delta Y_{i,t-2}$	past changes predict recent changes	ends at $t-2$: $\mathbb{E}[\Delta Y_{i,t-2}\Delta\varepsilon_{it}] = 0$	$t \geq 3$

- Apply 2SLS with one of the two as the instrument: just-identified, consistent as $N \to \infty$ with T fixed.
- **Its limits.** One instrument per equation, when at $t = 5$ the levels $Y_{i0}, \ldots, Y_{i3}$ are all valid; and plain 2SLS ignores that $\Delta\varepsilon_{it}$ and $\Delta\varepsilon_{i,t-1}$ share a shock. Consistent but imprecise, which is why GMM replaced it.

Difference GMM (Arellano and Bond 1991)

Step 1. Difference away μ_i: $\Delta Y_{it} = \alpha \Delta Y_{i,t-1} + \Delta X_{it}'\beta + \Delta \varepsilon_{it}$.

Step 2. Instrument $\Delta Y_{i,t-1}$ with lagged *levels*. Anderson–Hsiao used $Y_{i,t-2}$ alone; every earlier Y is just as **valid** (dated before any shock in $\Delta\varepsilon_{it}$) and still **relevant**, so use them all:

equation at	instruments for $\Delta Y_{i,t-1}$
$t = 2$	Y_{i0}
$t = 3$	$Y_{i0},\ Y_{i1}$
$t = 4$	$Y_{i0},\ Y_{i1},\ Y_{i2}$
t	$Y_{i0}, \ldots, Y_{i,t-2}$

$$Z_i = \begin{pmatrix} Y_{i0} & 0 & 0 & 0 & 0 & 0 \\ 0 & Y_{i0} & Y_{i1} & 0 & 0 & 0 \\ 0 & 0 & 0 & Y_{i0} & Y_{i1} & Y_{i2} \end{pmatrix} \begin{matrix} t=2 \\ t=3 \\ t=4 \end{matrix}$$

- Anderson–Hsiao had one condition per equation; now there are many more conditions than parameters. What do you *do* with them?

What GMM does with the moment conditions

Stack the instruments and the differenced residuals of firm i and average over firms:

$$\bar{g}(\alpha) \;=\; \frac{1}{N}\sum_{i=1}^{N} Z_i'\,\Delta\varepsilon_i(\alpha), \qquad \hat{\alpha} \;=\; \arg\min_{\alpha}\; J(\alpha) = N\,\bar{g}(\alpha)'\,W\,\bar{g}(\alpha).$$

- Each entry of $\bar{g}(\alpha)$ is the sample covariance between one instrument and the residual.
- **The Overidentification Problem:** Validity says the *population* covariances are zero. But with more instruments than parameters, no single α can zero out all the *sample* covariances at once.
- **The GMM Solution:** Pick the α that makes them *jointly as close to zero as possible*.
- Each instrument on its own gives a "just-identified" estimate. GMM is simply a weighted average of these estimates. The matrix W downweights the noisy, high-variance instruments (1-step vs 2-step GMM).

Weak instruments: when a level cannot predict a change

Difference GMM asks a *level* $Y_{i,t-2}$ to predict a *change* $\Delta Y_{i,t-1}$. But if

- $\alpha \to 1$: little mean reversion; where a firm *is* says nothing about where it is *going*.
- σ_μ^2 **large**: firms differ *permanently*, and permanent differences predict no change.

So $\mathrm{Cov}(\Delta Y_{i,t-1}, Y_{i,t-2}) = -\sigma_\varepsilon^2/(1+\alpha)$ stays bounded while
$\mathrm{Var}(Y_{i,t-2}) = \sigma_\mu^2/(1-\alpha)^2 + \sigma_\varepsilon^2/(1-\alpha^2)$ explodes; the first-stage correlation collapses:

$\sigma_\mu^2/\sigma_\varepsilon^2$	$\alpha = 0.5$	0.8	0.9	0.95	0.99
0 (no firm effect)	−0.50	−0.32	−0.22	−0.16	−0.07
1	−0.25	−0.10	−0.05	−0.02	−0.00
4	−0.14	−0.05	−0.03	−0.01	−0.00

- Bottom-right is finance: large standard errors, bias toward the within estimate.
- **Blundell and Bond's answer.** A change **always predicts a level**: swap roles.

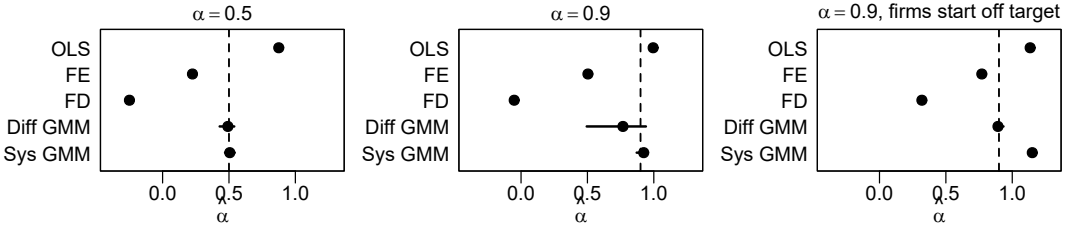
System GMM (Blundell and Bond 1998)

Keep the differenced equation and **bring back the levels equation**:

equation	regressor	instruments
differences: $\Delta Y_{it} = \alpha \Delta Y_{i,t-1} + \Delta \varepsilon_{it}$	$\Delta Y_{i,t-1}$	lagged *levels* $Y_{i,t-2}, Y_{i,t-3}, \ldots$
levels: $Y_{it} = \alpha Y_{i,t-1} + \mu_i + \varepsilon_{it}$	$Y_{i,t-1}$	lagged *differences* $\Delta Y_{i,t-1}$

- **Relevance does not fade.** The regressor contains the instrument, $Y_{i,t-1} = Y_{i,t-2} + \Delta Y_{i,t-1}$: a firm that just moved up is now higher, whatever α and σ_μ^2; no mean reversion needed.
- **Requires a new assumption:** The levels error still contains the firm fixed effect μ_i. We need $\text{Cov}(\Delta Y_{i,t-1}, \mu_i) = 0$.
  - A firm's recent change says nothing about its permanent level.
  - *It fails:* When firms are still converging to their steady state (e.g., IPOs, post-crisis panels, post-deregulation).
- **The trade:** Difference GMM needs *news* (low persistence). System GMM needs *settled firms* (steady state).

The simulation: five estimators in three worlds



500 firms, $T = 6$, 100 replications. Dots: medians; bars: 10th–90th percentiles; dashed: truth. OLS biased up, FE and FD down.

- $\alpha = 0.5$. Both GMM estimators sit on the truth (difference 0.49, system 0.51).
- $\alpha = 0.9$. Difference GMM wide and low (0.77): weak instruments. System GMM tight and right (0.92).
- Firms start away from steady state μ_i. Difference GMM recovers (0.89); system GMM lands at 1.15, above even OLS
- Report OLS and FE beside GMM; outside their range is a warning.

Summary

1. **Removing μ_i makes the lagged outcome endogenous.**
2. **Sequential exogeneity replaces strict exogeneity**: every Y dated $t-2$ or earlier is orthogonal to $\Delta\varepsilon_{it}$. Anderson–Hsiao uses one such lag by 2SLS; difference GMM uses them all and makes the sample moments as close to zero as W allows.
3. **The instruments are weak when the series is persistent or firm effects are large**. System GMM instruments the level with the change instead, at the price of mean stationarity, an assumption you must defend.
4. **In practice**: report OLS and FE beside the GMM column. OLS is biased up and FE down, so a GMM estimate outside their range signals a failed assumption. The diagnostics that test the assumptions directly, AR(2), Hansen and the instrument count, are in Roodman (2009) and Flannery and Hankins (2013).

References

- Anderson, T. W. and C. Hsiao (1981). Estimation of Dynamic Models with Error Components. *Journal of the American Statistical Association* 76, 598–606.
- Arellano, M. and S. Bond (1991). Some Tests of Specification for Panel Data: Monte Carlo Evidence and an Application to Employment Equations. *Review of Economic Studies* 58, 277–297.
- Blundell, R. and S. Bond (1998). Initial Conditions and Moment Restrictions in Dynamic Panel Data Models. *Journal of Econometrics* 87, 115–143.
- Flannery, M. and K. Hankins (2013). Estimating Dynamic Panel Models in Corporate Finance. *Journal of Corporate Finance* 19, 1–19.
- Lemmon, M., M. Roberts and J. Zender (2008). Back to the Beginning: Persistence and the Cross-Section of Corporate Capital Structure. *JF* 63, 1575–1608.
- Nickell, S. (1981). Biases in Dynamic Models with Fixed Effects. *Econometrica* 49, 1417–1426.
- Roodman, D. (2009). How to Do xtabond2: An Introduction to Difference and System GMM in Stata. *Stata Journal* 9, 86–136; and A Note on the Theme of Too Many Instruments. *Oxford Bulletin of Economics and Statistics* 71, 135–158.